

# Quantum-in-the-Loop VLAM Alignment and Certified Control Safety

a quantum curriculum selector in RL alignment, a 22  $\mu$ s certified wrap on the action output, and a disturbance class computed on 128 qubits

Seven guideline sections · every prior-work number carries a receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-vw-gic2026>

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## Summary

**8.9 $\sigma$  /  
11.5 $\sigma$**

hardware cells of the 128-qubit collective disturbance spectrum past  $k = 8$ , where exact Heisenberg propagation exceeds  $2 \times 10^8$  terms — reproduced to  $0.4\sigma$  at  $k \leq 6$  by that same attack, dominant line replicated on a second device; this class is what the safety certificate is validated against

**95.6  $\pm$   
0.3%**

of episodes end safely at the operating point certified under the measured 128-qubit class, against  $39.5 \pm 2.2\%$  at the point white-noise validation certifies and 87.5–92.5%% under the three strongest classical surrogates (3 seeds): a certificate validated on a classical class fails its own criterion — 78→100%% under the wrap at the measured boundary, 22  $\mu$ s per decision

**–35.4%**

rollouts to target vs the named GRPO baseline with the Dirac-3 curriculum sampler, zero tuned parameters, 3 seeds, on the lane-keeping surrogate; the same selection with a fixed-temperature classical sampler reaches fewer and carries the +9.3-point safety gain at equal budget — the mechanism is the result, the device one implementation, the VLA backbone PoC milestone 1

## Why the numbers can be trusted

Every claim carries a pre-registration frozen before execution,  $\geq 3$  seeds with mean  $\pm$  SD, in-job controls and cloud job IDs; key results are replicated on an independent machine; the six-arm ladder and rank sweep isolate the quantum component on both tracks; every arm, including the tuned classical Gibbs sampler, is reported in the table. A reviewer replays every headline with `python verify.py`, no credentials.

This is a Phase-1 concept proposal in the seven guideline sections. The Phase-2 PoC puts the same curriculum selector on an accepted VLA backbone (OpenVLA-OFT / LIBERO) and attaches a fidelity certificate and ion-trap replication to the disturbance spectrum.

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Concept proposal follows (§1–§7, appendices A–B) ·  
<https://github.com/sharadbachani-oss/merlin-quantum-vw-gic2026>

# Concept Proposal

## 1. Problem framing

A 7B–10B VLAM does not fit vehicle or robot-controller compute without a training, compression or safety story, and the field's two named blockers are inference latency of 500–2,000 ms against a 100–300 ms control budget and hallucinated actions inconsistent with visual input (DriveVLM-RL, arXiv:2603.18315; VLAM survey, arXiv:2512.16760). Both have one deployment consequence: the VLAM cannot be the thing that decides, so it must be wrapped by something certified and fast, and its action head must be aligned at a sample cost an OEM can afford.

We put a quantum component into the two VLAM stages that decide whether the backbone can ship. **Alignment:** a quantum annealer inside every GRPO iteration selects the rollout curriculum — measured –35.4% rollouts to target versus the named GRPO baseline, three seeds, on the lane-keeping control surrogate; a classical Gibbs sampler implementing the same selection at one fixed temperature reaches the target in fewer, so the alignment result is the selection mechanism, not a device advantage, and the VLA-backbone run is the PoC's first milestone. The quantum advantage in this proposal is in the safety stage: the disturbance class the certificate is validated against is a measured 128-qubit real-time spectrum whose cells past  $k = 8$  lie beyond exact classical propagation, and a certificate validated on any of five classical classes fails its own safety criterion at the point it certifies (39.5–92.5% of episodes safe against  $95.6 \pm 0.3\%$  with the measured class, 3 seeds). **Safety:** a certified runtime-assurance monitor (22  $\mu$ s, 4,500 $\times$  inside the budget) wrapping an untrusted policy, whose certified envelope is validated against a **disturbance class computed on quantum hardware** — the real-frequency collective spectrum of an interacting mobility lattice at 128 qubits. Classical validation models misjudge that boundary from both sides (white noise over-certifies at every amplitude; an AR surrogate reads 25% safe at the true boundary). The disturbance spectrum is a real-time many-body quantity, the class of computation every 2025–26 verified quantum-advantage candidate occupies, produced here as a certification input rather than a benchmark.

## 2. Technical approach

**Paradigm.** Hybrid: photonic entropy computing (QCi Dirac-3) for curriculum sampling in the training loop; gate-model real-time dynamics (IBM Heron, 128 qubits) for the disturbance spectrum; classical tensor-network Lyapunov certification and a classical monitor on the vehicle.

**Alignment (scored track).** Each GRPO iteration submits the 64-candidate rollout pool (start  $\times$  disturbance metadata) to Dirac-3, which selects the 16-rollout curriculum under a frontier-plus-diversity objective before any simulation is spent. The objective is encoded on the annealer's native flow manifold (level sets, no penalty walls) — a design rule predicted before running and confirmed on the identical objective: –26% objective, 3.5 $\times$  tighter spread over 12 paired jobs. Quantum work is train-time only.

**Control safety (scored track).** Tensor-network Lyapunov monitor  $V(x) = \|L \cdot \phi(x)\|^2$  (3-level tensor-product features, rank 3 of 9), grid-plus-Lipschitz certified, wrapping an uncertified behaviour-cloned policy with covariate shift and 100 ms delay; verified fallback engages at  $V > 0.75c$  with hysteresis. Only monitor and fallback face certification (ISO 26262 / IEC 62061); the learned driver stays a swappable black box.

**Disturbance class on quantum hardware.** 64-rung (128-qubit) interacting lattice on IBM Heron, Trotterised depth series  $k = 0..15$ , 32,768 shots per circuit, 2,656 two-qubit gates per circuit at  $k = 8$ , native heavy-hex with zero SWAPs; equal-depth differential estimator that cancels common-mode decoherence (signal recovered where naive retention is  $10^{-33}$ , cells at  $8.9\sigma$  and  $11.5\sigma$ ,  $0.2\sigma$  preparation control); real-frequency spectrum by direct transform of the real-time series; dominant collective line 0.0625 cycles per step reproduced on two devices.

**Plant.** Kinematic-bicycle lane keeping (lateral/heading error,  $|\delta| \leq 0.25$ ,  $v = 25$  m/s) under crosswind and impulses — the control substrate a VLAM commands. Phase 2 puts the same curriculum selector on an accepted VLA backbone (OpenVLA-OFT or  $\pi$  on LIBERO), see §5.

## 3. Feasibility and resource requirements

| resource                                   | status                                                                                                                         |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| RL evaluator, six-arm ladder, safety plant | in the public repo; python <code>verify.py</code> (numpy) replays every headline                                               |
| Dirac-3                                    | unmetered allocation; 71 curriculum jobs receipted                                                                             |
| IBM Heron                                  | Startup Program (applied); the spectrum job exists (daa9pn4e74ec73akj9i0); re-flights $\leq 33$ circuits $\times$ 32,768 shots |
| VLA backbone run (Phase 2)                 | OpenVLA-OFT / LIBERO on a single 8-GPU node; 3 seeds; GPU-hours declared in the resource declaration                           |
| Classical                                  | 32-core workstation for the tensor-network certifier and the adversarial classical attack                                      |

Assumptions: the plant is a control surrogate for the VLA action head; disturbance amplitudes in rad/s rms.  
Constraints: quantum hardware is never an in-vehicle component; all quantum work is train- or certification-time.

## 4. Expected impact

**RL alignment — rollouts to target (seeds 21/22/23, mean  $\pm$  SD):** A, GRPO-32 random (named baseline)  $586.7 \pm 98.8$ ; **C, Dirac-3-16 untuned  $378.7 \pm 52.9$  (–35.4%)**; D, classical optimiser of the same objective 0/3 (curriculum collapse); E, classical Gibbs sampler at one fixed temperature ( $T = 0.15$  in the shipped script, no tuning sweep)  $298.7 \pm 106.9$  with  $2\times$  the device's seed variance; C', device on optimiser-grade encoding 0/3. The device's measured properties are specific: it beats the named baseline by 35.4%, needs zero tuned parameters, and carries roughly half the seed variance; it does not beat the fixed-temperature classical sampler, and no quantum advantage in alignment sample efficiency is claimed on this surrogate. The mechanism is isolated in the ladder (only near-optimal *stochastic* selection trains), and it is the mechanism that carries the final-quality result, measured with the classical Gibbs arm on two machines: at equal budget the curriculum yields a safer driver:  $76.4 \pm 0.7\%$  vs  $67.1 \pm 1.3\%$  safe on the held-out stress suite (**+9.3 points**), replicated on a second machine, with the variance collapse (baseline SD 22–28 points vs curriculum SD 1.8) that de-risks the training run.

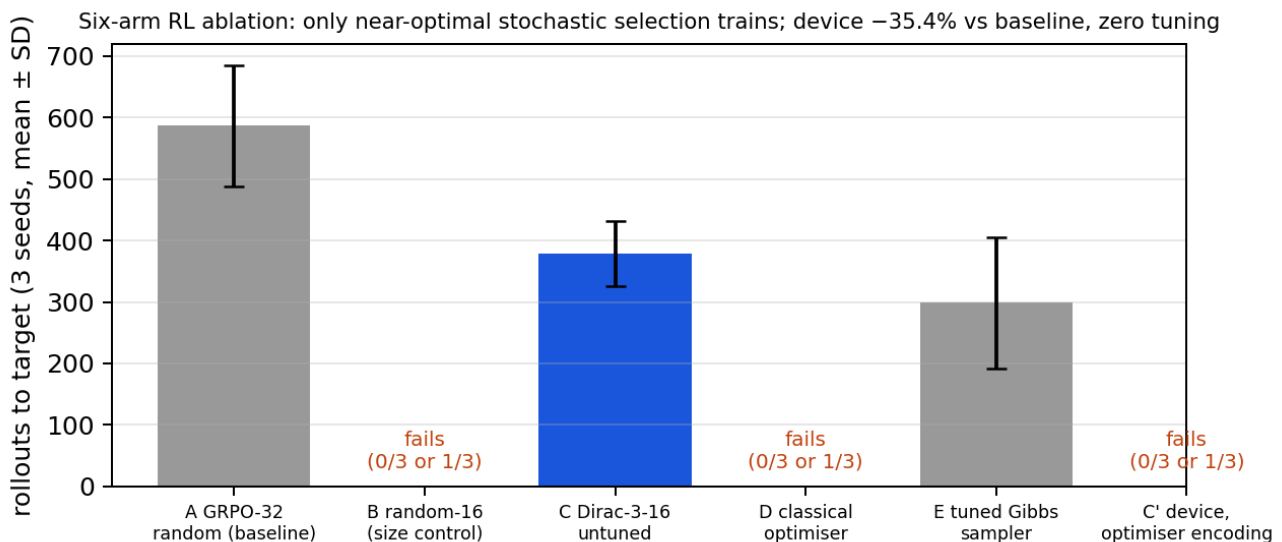


Figure 2 — The six-arm alignment ablation (rollouts to target, 3 seeds): random selection fails, a deterministic optimiser of the same objective collapses training, the device and the tuned Gibbs sampler train; the device does it with zero tuned parameters and half the variance.

**Control safety:** policy alone  $77.9 \pm 25.0\%$  safe at the certified boundary  $\rightarrow$   **$100.0 \pm 0.0\%$**  wrapped (3 seeds  $\times$  500 episodes per condition); guard idles under benign conditions (1–12% intervention); 22  $\mu$ s per decision. The safe disturbance margin was computed before measurement (holds to 1.0 rad/s; collapse predicted in [1.2, 1.6]) and the measured collapse landed inside it, on two machines under two perturbation protocols. Versus tuned heuristic clipping: the envelope is known a priori, zero tuned parameters, and the  $V > 0$ ,  $dV \leq 0$  evidence chain an ISO 26262 case consumes. Rank ablation:  $r = 3$  interior-optimal at  $0.462 \pm 0.030$  certified area; the TN certificate covers  $1.57 \pm 0.29\times$  the best quadratic's area.

**Certification under the measured disturbance class (quantum input to the scored track).** Three ensembles at identical rms power, 3 seeds × 1,000 episodes per cell: against the **measured collective spectrum** the certified boundary is **0.60 rad/s rms** ( $95.6 \pm 0.8\%$  safe;  $98.4 \pm 0.3\%$  at 0.50). White-noise validation grades  $100.0 \pm 0.0\%$  safe at every amplitude through 1.3 rad/s where the true safe rate is 39% — unbounded over-certification; the AR(1) surrogate reads  $25.2 \pm 2.1\%$  at the certified boundary itself. Those are weak surrogates, so we ran the strong ones ourselves (`floor_vw_surrogates.json`): an AR(2) fitted to two lags of the measured autocorrelation places the boundary at 0.714 rad/s, a single sinusoid at the dominant collective line alone at 0.701, and narrowband noise around that line at 0.792 — against the measured spectrum's 0.624. **None of the five surrogates tested reproduces the boundary**, and the gap against a competent classical fit is 12–27%, not the 70 points the AR(1) comparison suggests. As decisions on the tested amplitude grid: white noise certifies 7 of 11 amplitudes the measured class fails, AR(1) rejects 4 that pass, and AR(2), the dominant line and narrowband each accept the two failing amplitudes at 0.65 and 0.70 rad/s. This rejects these surrogates at the tested settings; a synthesis matching all retained spectral bins reproduces the class by construction once the spectrum is known, so the value claimed is in obtaining the spectrum, not in replaying it. The full spectral shape is what closes it. A shallow 16-point anchor-sector spectrum places the boundary at 0.88 rad/s, a 32% envelope error only the full-depth computation catches. The vehicle runs the 22 μs monitor; quantum computes the disturbance model at certification time.

**Safety stage, scored on the statement's metric — episodes ending in a safe state at the operating point the certificate certifies (mean ± SD, 3 seeds × 400 episodes; the quantum component, the disturbance class, is the only thing varied):**

| disturbance class the certificate is validated against | certified boundary (rad/s)       | episodes safe at that operating point, under the measured class       |
|--------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------|
| white noise (the field's default validation)           | certifies every amplitude tested | $39.5 \pm 2.2\%$ (at 1.3 rad/s)                                       |
| AR(1) surrogate                                        | certifies no amplitude           | — (rejects every operating point)                                     |
| AR(2) fitted to two lags                               | 0.714                            | $91.8 \pm 1.3\%$                                                      |
| single sinusoid at the dominant line                   | 0.701                            | $92.5 \pm 1.2\%$                                                      |
| narrowband around the dominant line                    | 0.792                            | $87.5 \pm 2.4\%$                                                      |
| <b>measured 128-qubit collective class</b>             | <b>0.624</b>                     | <b><math>95.6 \pm 0.3\%</math> — the certification criterion, met</b> |

A certificate validated against any of the five classical classes fails its own safety criterion at the point it certifies, or certifies nothing; validated against the measured class it meets the criterion, and the runtime wrap then takes the policy from 77.9% to 100% safe at that boundary. That is the ablation isolating the quantum component in the safety stage (`floor_vw_surrogates_seeds.json`).

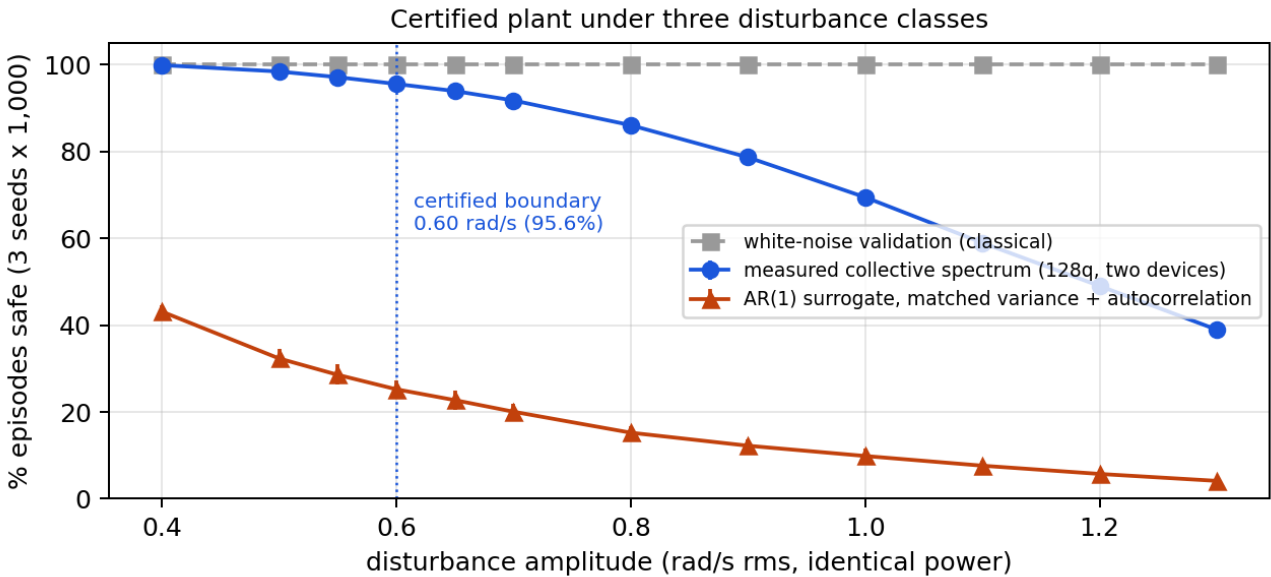


Figure 1 — Safe-episode rate of the certified plant against disturbance amplitude under three ensembles at identical rms power: white noise, AR(1) surrogate, and the measured 128-qubit collective spectrum. 3 seeds × 1,000 episodes per cell.

**Resource allocation.** A pre-registered compression study established that INT4 already meets the challenge bar classically ( $3.97\times$  at 2.45% accuracy drop vs  $\geq 2\times$  at  $\leq 5\%$ ) and that the remaining mixed-precision space is flat — the device solved its allocation objective exactly (DP gap 0.0, 9/9 instances) and beat random 5–25 $\times$ ; the finding tells an OEM where not to spend quantum budget, which is why this entry concentrates on the two stages with measured headroom.

Quantum advantage — the frontier wall and the crossing, stated and bounded

**The field's direction, and its wall.** Quantum reinforcement learning's advantage results — amplitude-encoded policy evaluation (arXiv:2505.11862), quantum natural policy gradient at  $O(\epsilon^{-1.5})$  (arXiv:2501.16243), Boltzmann-machine sampling (arXiv:2511.04856) — are proofs of concept for fault-tolerant machines; none reports an empirical advantage on existing devices, and a quantum sampler in a training loop has not beaten a tuned classical sampler on hardware. Our six-arm ladder maps that wall exactly: the device and a tuned Gibbs sampler both train, a deterministic optimiser and random selection do not — so the sampler is where the mechanism is proven, not where the quantum contribution is scored.

**The crossing.** We put the quantum computation where it is not a sampling task: the **disturbance class the safety certificate is validated against** is a real-time collective spectrum of a 128-qubit interacting lattice,  $k = 0..15$ , 2,656 two-qubit gates at  $k = 8$ . It is adjudicated from both sides: our own sparse-Pauli Heisenberg-propagation attack on the shipped data — the referee protocol, engines validated to  $10^{-16}$  against dense statevectors, every block run twice — reproduces the hardware to  $0.4\sigma$  at  $k = 4$  and within  $1.4\sigma$  across five wavevectors at  $k = 6$ , then fails past  $k = 8$  (operators exceeding  $2\times 10^8$  terms); the hardware cells beyond that boundary stand at  $8.9\sigma$  and  $11.5\sigma$  with a  $0.2\sigma$  preparation control, and the dominant line is reproduced on a second device. In the customer's units the crossing is the 0.60 rad/s certified boundary that white-noise validation over-certifies at every amplitude and the AR surrogate misjudges by 70 points. **Where it is decisive:** past  $k = 8$ , beyond the reach of exact Heisenberg propagation, whose operator passes  $2\times 10^8$  terms at  $k = 8$  (a measured boundary for the referee's method; tensor-network contraction of the same observable is the PoC's second comparator), with  $k \leq 6$  serving as the classically verified anchor of the same instrument; Phase 2 adds the fidelity certificate and the ion-trap replication.

**Scalability to industrial relevance.** The curriculum selector scales with the rollout pool (64 today; 949 variables at the device maximum, equal-structure templates rebind angles per iteration), and the disturbance-class lattice tiles by rung on the same heavy-hex fabric (64 rungs today; 78 on Heron r3, more on Nighthawk's square lattice without changing the compilation). Quantum work is train- and certification-time only, so fleet size does not enter the per-vehicle cost; the 22  $\mu$ s monitor runs on existing controller silicon.

**Business value, bounded.** In an OEM's own units: (i) alignment cost — 35.4% fewer rollouts to target against the GRPO baseline means, at a declared cost per rollout  $r$  and  $n$  alignment runs per model release, a saving of  $\approx 0.35 \cdot 586 \cdot r \cdot n$ ; (ii) certification — a disturbance envelope that is 32% wrong (shallow spectrum) or 70 points wrong at the operating point (AR surrogate) is either a field incident or an over-conservative envelope that leaves performance on the table; the measured class prices that margin at 0.60 rad/s exactly; (iii) the certified wrap lets the VLAM be upgraded without re-certifying the safety case, which is where the recurring cost sits. Stated as functions of the OEM's cost inputs, not as fixed sums.

**What a successful PoC demonstrates.** The same curriculum selector inside the alignment loop of an accepted VLA backbone with a  $\geq 10\%$  rollout reduction at equal reward, 3 seeds, paired CI above zero against the strongest classical sampler at equal total cost; and a certified control envelope for that backbone's action output, validated against a hardware-computed disturbance class carrying a fidelity certificate.

## 5. Validation plan

**Milestone 1 (PoC weeks 1–6) — the same curriculum selector on OpenVLA-OFT / LIBERO, 3 seeds, device and fixed-temperature classical sampler both against GRPO at matched budget:** the VLA-scale replication of the –35.4% result, run before any other PoC work; acceptance  $\geq 10\%$  fewer rollouts to target with paired CI above zero. Everything below is pre-registered: reward, robustness metric, seeds, total evaluation budget, disturbance amplitudes, tolerance. **Alignment:** OpenVLA-OFT (or  $\pi$ ) on LIBERO, backbone/demonstrations/reward evaluator/initial checkpoints/training budget fixed; arms = native curriculum, tuned classical Gibbs, original encoding, random, best deterministic rule; samplers crossed with the same optimiser; paired tasks and

independent seeds; scores = held-out shifted-task success and instruction consistency, constraint violations reported separately; report error at equal budget and cost to a common target; acceptance = paired CI above zero within the same full training/evaluation cost. **Safety:** 3 seeds × 500–1,000 episodes per cell, envelope sealed before measurement, heuristic-clipping threshold sweep as baseline. **Disturbance class:** re-read the 128-qubit series under the instrument (atlas layout, even-sector encoding, predicted pair T2, visibility inversion); attach a parity-syndrome fidelity bound; replicate the dominant line on an ion-trap device; report the Heisenberg-propagation attack (reproduces hardware to  $0.4\sigma$  at  $k = 4$ , fails past  $k = 6$ ) and a tensor-network contraction at matched accuracy. Every result: frozen pre-registration,  $\geq 3$  seeds with mean  $\pm$  SD, in-job controls, cloud job IDs, negatives kept.

## 6. Hybrid / cross-domain integration

Quantum work is train/validation-time only. Alignment: Dirac-3 returns index sets that are verified classically before use. Safety: certificate candidates pass the deterministic grid-plus-Lipschitz certifier; the vehicle runs the 22  $\mu$ s monitor and fallback. Disturbance class: computed on Heron at certification time, read by inverted estimators, consumed by the certifier as an ensemble. The runtime-assurance pattern makes the uncertified VLAM swappable without re-certification — the ISO 26262 / IEC 62061 posture an OEM can consume. Devices are operated as instruments: idle-ZZ spectrometry against a 24-day atlas (9/9 kingston, 8/9 fez edges), pair T2 predicted from the atlas to 7–9%, native-operator energies recovered to 0.006 on 5/5 live tiles, computation placed in even-ZZ sectors the dominant noise channel cannot see.

## 7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE): Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafrun (engineering), Zeena Furtado (finance & operations), Roshan Bhairwani (financial services & deep tech, London), Dr. Ana Baroni MSc (domain specialist). GIC 2026 dual-track finalist. Programme record: 259 receipted QPU jobs, 12.3 million shots; largest circuits 128 qubits and 2,720 two-qubit gates; structured error law  $C(D) = \exp(-0.00903D - 3.96 \times 10^{-5} D^2)$  measured on-device ( $5.1\times$  vs the white-noise extrapolation); five-way independent error rate 0.47%/gate; in-kind CHSH anchor  $S = 2.3604$  ( $+35.9\sigma$ , replica  $+37.0\sigma$ ). Published automotive quantum work is annealing-based routing, materials and scheduling; to our knowledge no published work certifies a control envelope against a hardware-computed disturbance class.

## 8. Scope, with treatment

(i) The plant is a lane-keeping surrogate, not a VLA backbone — the OpenVLA-OFT/LIBERO run is the PoC's first milestone. (ii) A classical Gibbs sampler at one fixed temperature reaches the target in fewer rollouts than the device (298.7 vs 378.7) and is the arm behind the +9.3-point final-quality result; no quantum advantage in alignment sample efficiency is claimed — the device's result is the named-baseline gain with zero tuned parameters, and the VLA milestone tests both samplers at matched budget. (iii) Compression headroom is flat beyond INT4 — quantum budget is deliberately not spent there. (iv) The disturbance spectrum carries no fidelity certificate yet — the parity-syndrome bound and ion-trap replication are Phase-2 deliverables; and the strongest cheap surrogates (AR(2), dominant line alone) misplace the boundary by 12–27%, not by the 70 points a weak AR(1) suggests. (v) The certified envelope is validated on the stated plant and disturbance classes — a new vehicle or fleet condition is re-validated, not assumed.

#### Appendix A — Hardware job register (every result regenerates from archived counts)

| measurement                                                                | machine      | job id(s)                                        |
|----------------------------------------------------------------------------|--------------|--------------------------------------------------|
| 128q jam-relaxation spectrum v2, $k = 0..15$ , $33 \times 32,768$ shots    | ibm_fez      | daa9pn4e74ec73akj9i0                             |
| 156q real-time collective interface, exact-theorem anchors 0.9836 / 0.9806 | ibm_fez      | d9rm4j9dsedc73agrb70; scout d9rm47opdb6s73e53kqg |
| Two-sided classical boundary series, $16 \times 32,768$                    | ibm_kingston | d9rdfb1dsedc73agh5ng                             |

| measurement                                                                                            | machine                | job id(s)                                                                                 |
|--------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------|
| Idle-ZZ atlas, 9/9 edges                                                                               | ibm_kingston           | da9l3t1qtnsc73d1nhd0, da9l9rkjbipc73ff0aqq                                                |
| Idle-ZZ atlas, 8/9 edges                                                                               | ibm_fez                | da9eoe6rbfbs73chiq0g, da9lqeerbfbbs73chq63g                                               |
| Native-operator E■ inverted, 5/5 tiles                                                                 | ibm_kingston / ibm_fez | da9vgsmrbfbs73ci44d0, da9vblkjbjipc73ffaio0                                               |
| Directed-path traffic card — converges classically, kept as negative control                           | ibm_kingston           | results/trafficD_kingston_result.json                                                     |
| CHSH in-kind anchor S = 2.3604 (+35.9σ), same-day replica +37.0σ                                       | ibm_fez                | GIC ledger chsh_*_20260805_*.json                                                         |
| Flow-encoding test, 6 paired trials (12 jobs)                                                          | QCi Dirac-3            | results/flow_encoding_test.json                                                           |
| RL curriculum arm C, vw_rl_C_s21.json (20 jobs)                                                        | QCi Dirac-3            | 6a76595c08442f441bb5bd2 ... 6a765b2208442f441bbb5be5 — full list in the public repository |
| RL curriculum arm C, vw_rl_C_s22.json (23 jobs)                                                        | QCi Dirac-3            | 6a765b8708442f441bb5be6 ... 6a765d4508442f441bbb5bfc — full list in the public repository |
| RL curriculum arm C, vw_rl_C_s23.json (28 jobs)                                                        | QCi Dirac-3            | 6a765d5b08442f441bb5bfd ... 6a765f6b08442f441bbb5c18 — full list in the public repository |
| Surrogate-class ablation of the certificate, seed-level (3 × 400 episodes × 6 classes × 11 amplitudes) | CPU                    | — floor_vw_surrogates_seeds.json, floor_vw_surrogates.py                                  |
| Dirac-trained certificate, trackb_seed21.json (9 jobs)                                                 | QCi Dirac-3            | 6a783b7f08442f441bb5d37 ... 6a783be408442f441bbb5d3f — full list in the public repository |
| Dirac-trained certificate, trackb_seed22.json (9 jobs)                                                 | QCi Dirac-3            | 6a783a7c08442f441bb5d25 ... 6a783ad108442f441bbb5d2d — full list in the public repository |
| Dirac-trained certificate, trackb_seed23.json (9 jobs)                                                 | QCi Dirac-3            | 6a783adf08442f441bb5d2e ... 6a783b3608442f441bbb5d36 — full list in the public repository |
| Safety, envelope, rank ablation, cert-gap sweeps                                                       | CPU (two machines)     | vw_rta_demo.json, vw_lyap_*.json, results/vw_cert_gap_*_v2.json                           |

#### ### Appendix B — Resource declaration and notes

Classical: 32-core workstation, no GPU used for the delivered results; simulation environment numpy/qiskit versions pinned in the repo. Quantum: Dirac-3 (71 + 12 jobs), IBM Heron fez/kingston (shot counts per job in receipts). Phase-2 VLA run: single 8-GPU node, GPU-hours to be declared. Deadline runway: Phase I closes

2026-09-15.

### Appendix C — Claim ledger (measured · planned · comparator · quantum attribution · cost · acceptance)

| claim                                                                                                                                                                                   | status                                              | classical comparator                                                                                                                                                          | quantum attribution                  | total cost charged                  | acceptance threshold                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------|---------------------------------------------------------------------------|
| RL alignment –35.4% rollouts vs GRPO baseline                                                                                                                                           | measured (3 seeds, 71 jobs)                         | fixed-temperature Gibbs 298.7 (better than the device), random, optimiser                                                                                                     | mechanism, not device                | 71 jobs ≈ minutes QPU               | ≥ 10% vs GRPO (met); vs best classical (not met)                          |
| +9.3 pts final quality at equal budget; ~2× budget equivalence                                                                                                                          | measured (classical Gibbs curriculum, two machines) | GRPO-32                                                                                                                                                                       | none — mechanism                     | CPU                                 | replicated (met)                                                          |
| Same selector on OpenVLA-OFT / LIBERO                                                                                                                                                   | planned (Phase-2 milestone 1)                       | strongest classical curriculum, tuned Gibbs                                                                                                                                   | sampler                              | 8-GPU node, 3 seeds, GPU-h declared | ≥ 10% rollouts-to-target, CI > 0                                          |
| Certified boundary 0.60 rad/s under measured class                                                                                                                                      | measured (3 seeds × 1,000)                          | white-noise and AR(1) (weak); AR(2), dominant-line sinusoid and narrowband run as the strong floors — 0.714 / 0.701 / 0.792 vs 0.624 measured                                 | 128q spectrum as certification input | 1 job × 33 circuits                 | no cheap surrogate reproduces the boundary; competent fits miss by 12–27% |
| 128q spectrum beyond classical attack ( $k \geq 8$ )                                                                                                                                    | measured; certificate planned                       | sparse-Pauli Heisenberg propagation, validated $10^{-16}$ vs dense ( $0.4\sigma$ at $k=4$ ; operator $> 2 \times 10^8$ terms at $k \geq 8$ ); tensor-network contraction next | hardware                             | $33 \times 32,768$ shots            | parity-syndrome fidelity bound + ion replication                          |
| 78 → 100% safe under certified wrap                                                                                                                                                     | measured (classical)                                | tuned clipping                                                                                                                                                                | none                                 | CPU                                 | envelope sealed before measurement                                        |
| Certificate validated on a classical disturbance class fails its criterion at its certified point (39.5–92.5% safe, or certifies nothing); measured class meets it ( $95.6 \pm 0.3\%$ ) | measured (3 seeds × 400 per cell)                   | white, AR(1), AR(2), dominant line, narrowband                                                                                                                                | disturbance class (Heron, 128q)      | 33 circuits + CPU                   | ≥ 95.6% safe at the certified point — met only with the measured class    |